Radhikadua123
Google Scholar

Radhika Dua

Personal Website

radhikadua@nyu.edu
linkedin.com/in/duaradhika/

Research Interests

Reliable and clinically grounded medical AI for real-world clinical deployment: interpretable evaluation of medical generative models, uncertainty quantification for safe decision-making, and cross-modal learning to extend access to advanced imaging.

Education

New York University, New York, United States *2024-Present*
PhD, Center for Data Science CGPA: **3.98/4.0**

- Advisors: Prof. Eric Oermann and Prof. Kyunghyun Cho.
- Affiliations: CDS, CILVR, ML2.
- **Google PhD Fellowship in Machine Learning & ML Foundations, NYU Center for Data Science Fellowship, NYU Center for Data Science Medical Fellowship**.

KAIST, Republic of Korea *2020-2022*
Masters, Graduate School of AI CGPA: **4.05/4.3**

- **Outstanding Performance**
- Thesis Topic: Task agnostic and post-hoc unseen distribution detection.
- Advisor: Prof. Edward Choi.
- KAIST International Students Scholarship

Panjab University, Chandigarh, India *2015-2019*
Bachelor of Engineering With Honours, Information Technology **CGPA: 9.24/10**

- **Third position** in the department.

Research Experience

Google DeepMind / Google Research *Predoctoral Researcher* *Sept 2022 - Aug 2024*

- Mentors: Dr. Gaurav Aggarwal, Alok Talekar, Ishan Deshpande, Dr. Sujoy Paul
- Agricultural Landscape Understanding (ALU)
  - Led the **ALU Project**, which aims to segment agriculture-related feature types by utilizing satellite imagery.
  - Enhanced data handling by addressing **class imbalance**, **scarcity of unlabeled data**, and **domain generalization** challenges due to geographic and temporal distribution shifts.
  - The project was presented in KDD 2026, and in AGU 2023 as an oral presentation, Google Research Conference 2023, Google I/O 2023 [demo], Google I/O Connect India 2023 [video], Google For India 2022 [video, G4I blog, google blog 2023, google blog 2024, google blog 2025].
  - **Launched in 11 countries** in global south via API https://agri.withgoogle.com/
- One Shot Segmentation
  - Developed an innovative segmentation technique capable of **learning effectively from a single labeled instance**.
  - Objective is to **circumvent the conventional model training process** for segmentation tasks, which typically requires the annotation of extensive datasets.
  - Performance **comparable to Visual Prompting methods** that are trained on tasks relevant for segmentation.

New York University *PhD Researcher* *Aug 2024 - Present*

- Mentors: Prof. Eric Oermann, Prof. Kyunghyun Cho
- Clinically-Grounded Report Evaluation (ICARE)
  - Developed **ICARE**, an interpretable and clinically-grounded agent-based evaluation framework for radiology report generation, aligning with **expert radiologist judgment** better than prior metrics and advancing safe deployment of report generation models.
  - Accepted to NeurIPS 2025 GenAI4Health and WiML Workshops.

- Built an interactive ICARE demo on NVIDIA DGX Spark early-access GPUs for real-time evaluation of medical text generation systems [Nvidia News] .
- Deployed at NYU Langone Health and is running on diverse patient population.

- Cross-Modal Diagnostic Modeling (Thoracle)
  - Leading **paired CXR–CT representation learning** to enhance diagnostic reasoning by supervising models across complementary imaging modalities.
- Monotonic Uncertainty Regularization (MUR)
  - Investigating **monotonic uncertainty principles** to ensure confidence improves with information, advancing reliability in medical AI systems.

EdLab, KAIST *Student Researcher* *Aug 2020 - Aug 2022*

- Mentor: Prof. Edward Choi
- Reliability of Deep Neural Networks
  - Developed a simple yet effective, task-agnostic and post-hoc unseen distribution detection method, which was accepted at **WACV 2023** and the **ARROW workshop in ECCV 2022**.
  - Designed a novel task for detecting attribute-based shifts, developed a dedicated dataset, and formulated an innovative approach, **currently under review at an A* conference**.
- ML for Healthcare
  - Investigated Federated Learning's practical applications in healthcare, which was accepted at **CHIL 2023**.
  - Proposed a method for detecting noisy ECG signals, enhancing precision in cardiology, presented at **ICPR 2022**.

Brown University *Summer Intern* *July 2020 - Sept 2020*

- Mentors: Prof. Srinath Sridhar (Brown Univ) and Prof. Leonidas Guibas (Stanford Univ).
- Canonicalization of 3D Pose
  - Conducted research on self-supervised canonicalization of 3D pose for full and partial shapes.
  - Paper accepted for a poster presentation to **CVPR 2022**.

Indian Institute of Technology Hyderabad, India *Visiting Researcher* *Jan 2019 - June 2020*

- Mentor: Prof. Vineeth N Balasubramanian
- ViQAR: Visual Question Answering and Reasoning
  - Conducted research in Vision and Language applications and introduced a new task, ViQAR: Visual Question Answering and Reasoning, which focuses on automatic generation of the answer, and of a rationale, given a visual query.
  - Paper accepted for a poster presentation to **MULA workshop, CVPR 2021**.

Celestini Project India *Research Intern* *June 2018 - Aug 2018*

- **Mentors:** Dr. Aakanksha Chowdhery (Google Brain and Tensorflow) and Prof. Brejesh Lall (IIT Delhi).
- Real-time Air Pollution Forecasting
  - Developed a temporal forecasting solution based on the historical data reported by Central Pollution Control board to predict the real-time and fine-grained air quality information in five locations of Delhi.
  - Blog post, Indian press and Demo youtube video.
  - Paper accepted for a poster presentation to **GlobalSIP 2019**.

Honors, Fellowships, and Awards

- **Google PhD Fellowship in Machine Learning & ML Foundations**, 2026–Present.
- **NYU Data Science Graduate Fellowship**, 2024–2029.
- **NYU Center for Data Science Medical Fellowship**, 2024–Present.
- **Travel Award, NeurIPS (Women in Machine Learning)**, 2025.
- **KAIST International Students Scholarship**, 2020–2022.
- **Grace Hopper Celebration India (GHCI) Student Scholarship**, 2018.
- **Celestini Project India (Marconi Society and Google)**, Second Prize, 2018.
- **Economic Times Campus Stars**, recognizing India's brightest engineering students, 2018–2019.
- **Hackathons:** Second Prize, Hack In The North 2017; Second Prize, Hack Infinity 2017; Sixth Prize, IndiaHacks 2017 by HackerEarth.

Publications

Conference Publications

- **Radhika Dua**, Young Joon (Fred) Kwon, Varun Yerram, Eric Karl Oermann, Kyunghyun Cho. **More Information, Less Uncertainty? Monotonic Uncertainty Regularization Under Progressive Information**, [Under review at ***ICLR 2027***].
- **Radhika Dua**, Young Joon (Fred) Kwon, Siddhant Dogra, Daniel Freedman, Diana Ruan, Motaz Nashawaty, Danielle Rigau, Daniel Alexander Alber, Kang Zhang, Kyunghyun Cho, Eric Karl Oermann, **Clinically Grounded Agent-based Report Evaluation: An Interpretable Metric for Radiology Report Generation**, [Under review at ***Nature Communications***] [paper]
- **Radhika Dua***, Frederick Wolff*, Kyunghyun Cho, Eric Karl Oermann. **Discharge Summary Generation Is Not Just Summarization: A Reality Check for Evaluation**, [Under review at ***ML4H 2026***].
- Lavender Y. Jiang, Angelica Chen, Xu Han, Xujin Chris Liu, **Radhika Dua**, Kevin Eaton, Frederick Wolff, Robert Steele, Jeff Zhang, Anton Alyakin, Qingkai Pan, Yanbing Chen, Karl L. Sangwon, Daniel A. Alber, Jaden Stryker, Jin Vivian Lee, Yindalon Aphinyanaphongs, Kyunghyun Cho, Eric Karl Oermann, **Generalist Foundation Models Are Not Clinical Enough for Hospital Operations**. [Under review at ***Nature Medicine***] [paper]
- **Radhika Dua**, Nikita Saxena, Aditi Agarwal, Alex Wilson, Gaurav Singh, Hoang Tran, Ishan Deshpande, Amandeep Kaur, Gaurav Aggarwal, ... , Alok Talekar, **Agricultural Landscape Understanding At Country-Scale**, ***KDD 2026***. [paper]
- **Radhika Dua**, Seongjun Yang, Yixuan Li, Edward Choi. **TAPUDD: Task Agnostic and Post-hoc Unseen Distribution Detection**, ***WACV 2023***. [paper]
- Seongjun Yang, Hyeonji Hwang, Daeyoung Kim, **Radhika Dua**, Jongyeup Kim, Eunho Yang, Edward Choi **Towards a Practical Utility of Federated Learning in the Medical Domain**, ***Conference on Health, Inference, and Learning (CHIL) 2023***. [paper]
- Rahul Sajnani, Adrien Poulenard, Jivitesh Jain, **Radhika Dua**, Leonidas Guibas, Srinath Sridhar. **ConDor: Self-Supervised Canonicalization of 3D Pose for Partial Shapes**, ***CVPR 2022***. [paper, website]
- Taehee Kim, ChaeHun Park, Jimin Hong, **Radhika Dua**, Edward Choi, Jaegul Choo. **Reweighting Strategy based on Synthetic Data Identification for Sentence Similarity Comparison**, ***COLING 2022***. [paper]
- **Radhika Dua***, Divyam Madaan*, Prerana Mukherjee, Brejesh Lall. **VayuAnukulani: Adaptive Memory Networks for Air Pollution Forecasting**, ***GlobalSIP2019***. [code, paper, slides]

Workshops

- **Radhika Dua**, Fred Kwon, Varun Yerram, Eric Oermann, Kyunghyun Cho. **More Information, Less Uncertainty? Monotonic Uncertainty Regularization Under Progressive Information**, ***Women in Machine Learning (WiML) Workshop, NeurIPS 2026*** and ***Grace Hopper Celebration (GHC) 2026***.
- **Radhika Dua**, Young Joon (Fred) Kwon, Siddhant Dogra, Daniel Freedman, Diana Ruan, Motaz Nashawaty, Danielle Rigau, Daniel Alexander Alber, Kang Zhang, Kyunghyun Cho, Eric Karl Oermann, **Clinically Grounded Agent-based Report Evaluation: An Interpretable Metric for Radiology Report Generation**, ***GenAI4Health Workshop*** and ***WIML Workshop, NeurIPS 2025***. [paper]
- **Radhika Dua**, Nikita Saxena, Aditi Agarwal, Alex Wilson, Gaurav Singh, Hoang Tran, Ishan Deshpande, Amandeep Kaur, Gaurav Aggarwal, ... , Alok Talekar, **Agricultural Landscape Understanding At Country-Scale**, ***Tackling Climate Change with Machine Learning Workshop, NeurIPS 2025***. [paper]
- **Radhika Dua**, Seongjun Yang, Yixuan Li, Edward Choi. **TAPUDD: Task Agnostic and Post-hoc Unseen Distribution Detection**, ***ARROW Workshop, ECCV 2022***. [paper]

* denotes equal contribution

- **Radhika Dua**, Jiyoung Lee, Joon-myoung Kwon, Edward Choi. **Automatic Detection of Noisy Electrocardiogram Signals**, ***International Workshop on Pattern Recognition in Healthcare Analytics, ICPR 2022***. [paper]
- **Radhika Dua***, Sai Srinivas*, Vineeth N Balasubramanian. **Beyond VQA: Generating Multi-word Answer and Rationale to Visual Questions**, ***MULA Workshop, CVPR 2021***. [paper, video, slides, poster]

Abstract Presentations

- **Radhika Dua**, Alex Wilson, Shubhika Garg, Gaurav Singh, Alok Talekar, Gaurav Aggarwal, Gopi Prashanth Ramineni, Bindiya Kurle. **Agricultural Landscape Understanding At Scale**, ***American Geophysical Union, AGU 2023***. [**Oral presentation**] [abstract]
- Alok Talekar, Aravindan Krishnan, **Radhika Dua**, Alex Wilson, Gopi Prashanth Ramineni, Arnab Basu. **Policy Driven Growth In Water Infrastructure By SmallHolder Farmers**, ***American Geophysical Union, AGU 2023***. [**Lightning presentation**] [abstract]
- Ishan Deshpande, Bhartendu Pandey, **Radhika Dua**, Amandeep Kaur, Alok Talekar, Anu Ramaswami. **AgroForestry in India - Towards Country Wide Finescale Carbon Accounting**, ***American Geophysical Union, AGU 2023***. [**Poster presentation**] [abstract]
- Karan Choudhary, Chintapalli Srirama Srirama Murthy, Sunil Kumar Dubey, **Radhika Dua**, Dinesh Tewari, Alok Talekar. **Crop analytics at field parcel level using multi-resolution satellite images, segmentation and mapping techniques: Towards a transformative solution for crop surveys**, ***American Geophysical Union, AGU 2023***. [**Poster presentation**] [abstract]
- Chintapalli Srirama Srirama Murthy, Sunil Kumar Dubey, Karan Choudhary, **Radhika Dua**, Ishan Deshpande, Alok Talekar. **Mapping tree cover over farm parcels and assessing the scope for agro-forestry development using high resolution satellite images and segmentation techniques**, ***American Geophysical Union, AGU 2023***. [**Poster presentation**] [abstract]

Preprints

- **Radhika Dua***, Jeonghoon Park*, Jimin Hong*, Daehoon Gwak, Yixuan Li, Jaegul Choo, Edward Choi. **Natural Attribute-based Shift Detection**, ***Arxiv 2021***. [paper]

Teaching Experience

- **Fundamentals of Machine Learning**, Section Leader and Grader, New York University, Fall 2026. Course instructor: Prof. Kyunghyun Cho.
- **Responsible Data Science (UA 202)**, Teaching Assistant, New York University, Spring 2025. Course instructor: Prof. Umang Bhatt.
- **Scientific Writing (CC500)** and **Graduate English Presentations (HSS583)**, Teaching Assistant, KAIST. Course instructor: Prof. Mik Fanguy.
- **Deep Learning for Computer Vision**, Teaching Assistant, NPTEL. Course instructor: Prof. Vineeth N. Balasubramanian.
- **Deep Learning for Computer Vision (CS5370)**, Teaching Assistant, IIT Hyderabad. Course instructor: Prof. Vineeth N. Balasubramanian.
- **Advanced Topics in Machine Learning (CS6360)**, Teaching Assistant, IIT Hyderabad. Course instructor: Prof. Vineeth N. Balasubramanian.

Talks and Presentations

- **Research Talks and Presentations**
  - *More Information, Less Uncertainty? Monotonic Uncertainty Regularization Under Progressive Information*
    - Samsung Research (Internal Research Talk), June 2026
  - *Clinically Grounded Agent-based Report Evaluation: An Interpretable Metric for Radiology Report Generation*
    - Global AI Frontiers Lab (GAIFL) Workshop, South Korea (Lightning Talk), July 2026
    - Global AI Frontiers Lab, Oct. 2025
    - AI Mixers, NYU Langone Health (Lightning Talk), 2025
  - *Agricultural Landscape Understanding at Scale*
    - Google Research Week (Lightning Talk), 2024
    - AGU (Oral Talk), 2023
    - Google Booth, AGU, 2023
  - *Towards Sustainable Agriculture Prioritizing the Global South Using Machine Learning*
    - IISc Open Day (Keynote Talk), 2023
- **Outreach and Student Sessions**
  - **Research Lightning Talk**, **NYU Center for Data Science PhD Admissions Information Session**, Oct. 2026 — presented medical AI research to prospective PhD students.
  - **CodeChef Campus Chapter UIET**, Speaker, 2016–2017 — programming sessions for junior-year students.
  - **Software Freedom Day**, Panjab University, Speaker, 2017 — encouraged junior-year students to contribute to open source and pursue STEM education.

Professional Activities

- **Organizing and Leadership**
  - **NYU CILVR Seminar Series**: Co-organizer (Spring 2025, Fall 2025, Spring 2026).
  - **WiML Workshop at NeurIPS 2026**: Travel-Funding Assistant SuperVolunteer.
- **Peer Reviewing**
  - **Conferences (Reviewer / Subreviewer):** ICLR (2027); CVPR (2022, 2025, 2026); ECCV (2022, 2024, 2026); ICCV (2023); NeurIPS (2025); BMVC (2026); WACV (2026); AAAI (2024); ICVGIP (2024); SDM (2020); ML4Health (2026).
  - **Workshops (Reviewer / Subreviewer):** WiML and GenAI4Health, NeurIPS (2026); WiCV, ECCV (2024).
  - **Scholarship Reviewing:** Grace Hopper Celebration (GHC) Scholarship (2020).
- **Mentoring**
  - Mentor, **Climate Tech Startup Meet 2023** by Google.
  - Poster Mentor, **Women in Machine Learning (WiML) Workshop**, NeurIPS 2020.
  - Mentor, **Hacksprint Version 2 Hackathon**, Panjab University, Chandigarh.
  - Fellow Campus Ambassador, **Campus Geek Ambassador Program by GeeksforGeeks**, 2017–2018.
- **Volunteering**
  - **Women in Machine Learning**, NeurIPS 2025.
  - **Google DeepMind Booth**, ICML 2024.
  - **Google Networking Event** and **Google Booth**, ICCV 2023.
  - Virtual conferences: **ICML 2021**, **ICML 2020**, and **ACL 2020**.